

Project Development Phase  
Performance Test

|               |                        |
|---------------|------------------------|
| Date          | 20 February 2026       |
| Team ID       | LTVIP2026TMIDS90550    |
| Project Name  | Heart Disease Analysis |
| Maximum Marks |                        |

Model Performance Testing:

DATA PREPROCESSING

| S.No | Parameter         | Description                              |
|------|-------------------|------------------------------------------|
| 1    | Missing Values    | Identified and handled                   |
| 2    | Data Types        | Verified (String, Number, Boolean)       |
| 3    | Outliers          | Checked and validated                    |
| 4    | Calculated Fields | Created derived fields                   |
| 5    | Grouping          | Created groups for categorical variables |
| 6    | Binning           | Created age groups and BMI categories    |
| 7    | Data Cleaning     | Removed duplicates and errors            |
| 8    | Validation        | Cross-checked data accuracy              |

UTILIZATION OF FILTERS

| S.No | Parameter           | Description                    |
|------|---------------------|--------------------------------|
| 1    | Age Filter          | Filter by age groups           |
| 2    | Gender Filter       | Filter by Male/Female          |
| 3    | Race Filter         | Filter by race categories      |
| 4    | Smoking Status      | Filter by smoker type          |
| 5    | Alcohol Consumption | Filter by drinking level       |
| 6    | Physical Activity   | Filter by activity level       |
| 7    | Diabetic Status     | Filter by diabetes condition   |
| 8    | Health Conditions   | Filter by comorbidity          |
| 9    | Filter Type         | Multi-select and single-select |
| 10   | Filter Behavior     | Real-time update on selection  |

---

# CALCULATION FIELDS USED

| S.No | Parameter               | Description                                              |
|------|-------------------------|----------------------------------------------------------|
| 1    | Age Groups              | Binned continuous age into categories                    |
| 2    | BMI Categories          | Created Underweight/Normal/Overweight/Obese              |
| 3    | Risk Score              | Calculated combined risk level                           |
| 4    | Prevalence Rate         | $(\text{Heart Disease Cases} / \text{Total}) \times 100$ |
| 5    | Percentage Distribution | Calculated % for each category                           |
| 6    | Count Measures          | Count of records per category                            |
| 7    | Yes/No Flags            | Converted boolean to readable text                       |
| 8    | Group Combinations      | Combined fields for analysis                             |

---

# DASHBOARD DESIGN

| S.No | Parameter        | Description                                                    |
|------|------------------|----------------------------------------------------------------|
| 1    | Dashboard 1      | Demographic Analysis (Gender, Age, Race)                       |
| 2    | Dashboard 2      | Lifestyle & Risk Factors (Smoking, Alcohol, Physical Activity) |
| 3    | Dashboard 3      | Overall Analysis (Summary KPIs, Combined View)                 |
| 4    | Layout           | Grid-based responsive layout                                   |
| 5    | Color Scheme     | Consistent color palette                                       |
| 6    | Navigation       | Tab-based navigation between dashboards                        |
| 7    | Visual Hierarchy | KPIs at top, detailed charts below                             |
| 8    | Interactivity    | Filters, tooltips, drill-down enabled                          |
| 9    | Accessibility    | Clear labels and readable fonts                                |
| 10   | Branding         | Project title and logo placeholders                            |

---

# STORY DESIGN

| S.No | Parameter      | Description                                  |
|------|----------------|----------------------------------------------|
| 1    | Story Point 1  | Demographic Analysis                         |
| 2    | Story Point 2  | Life style and Risk Factors                  |
| 3    | Story Point 3  | Overall Analysis                             |
|      |                |                                              |
|      |                |                                              |
|      |                |                                              |
| 4    | Narrative      | Each point includes description              |
| 5    | Visual Support | Charts accompany each point                  |
| 6    | Flow           | Logical progression from problem to solution |
| 7    | Export         | Can be exported as PDF/Image                 |